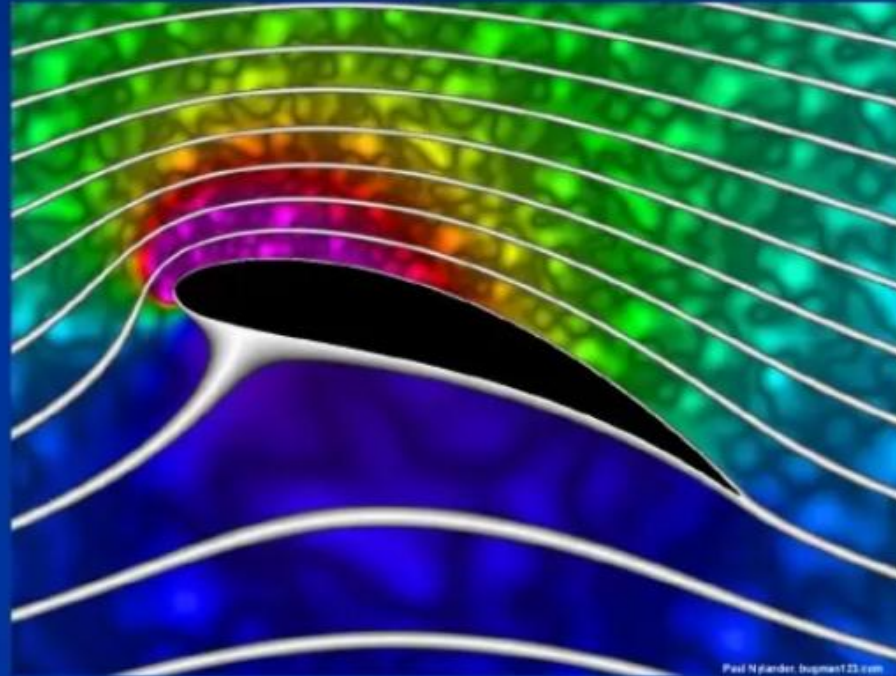


Aerodynamic Characteristics of a NACA 4412 Airfoil



Presented By: David Heffley
Mentor: Dr. Van Treuren
Scholar's Day
January 26, 2007



Overview

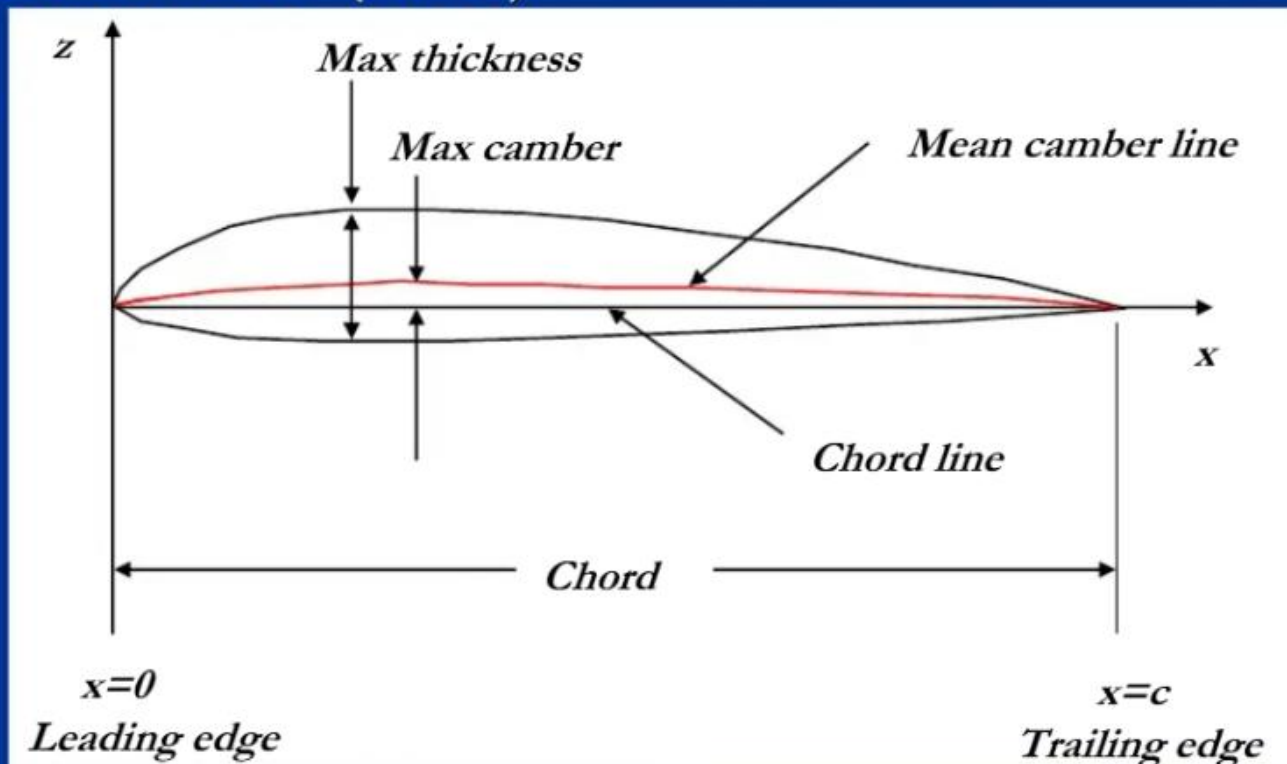
- Objective
- Theory
- Apparatus
- Experimental Comparison
- Results
- Summary
- Recommendations

Objective

- Study the lift and drag forces on a NACA 4412 airfoil
- Resolve discrepancy in wind tunnel data
- Develop experimental techniques for an airfoil
- Compare wind tunnel data
 - Force Balance to Pressure Distribution
 - Baylor data to published NACA data

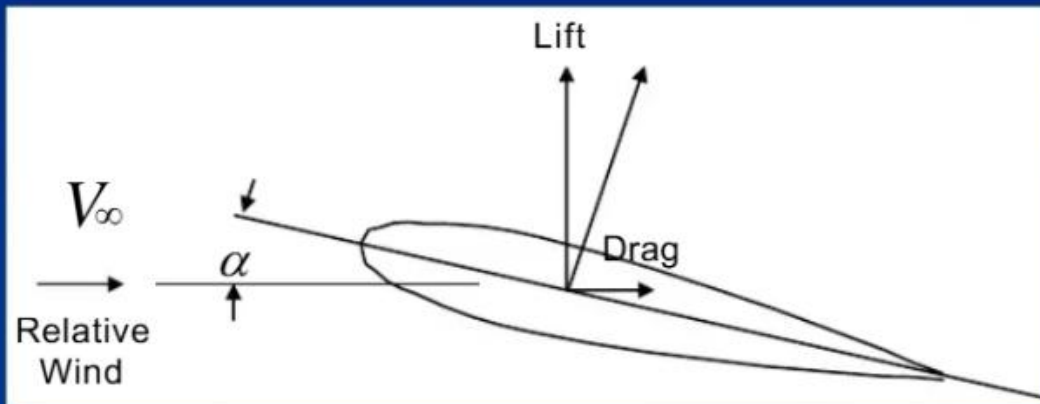
NACA 4412 Airfoil

- 4 digit code used to describe airfoil shapes
- 1st digit - maximum camber in percent chord
- 2nd digit - location of maximum camber along chord line (from leading edge) in tenths of chord
- 3rd and 4th digits - maximum thickness in percent chord
- NACA 4412 with a chord of 6"
 - Max camber: 0.24" (4% x 6")
 - Location of max camber: 2.4" aft of leading edge (0.4 x 6")
 - Max thickness: 0.72" (12% x 6")

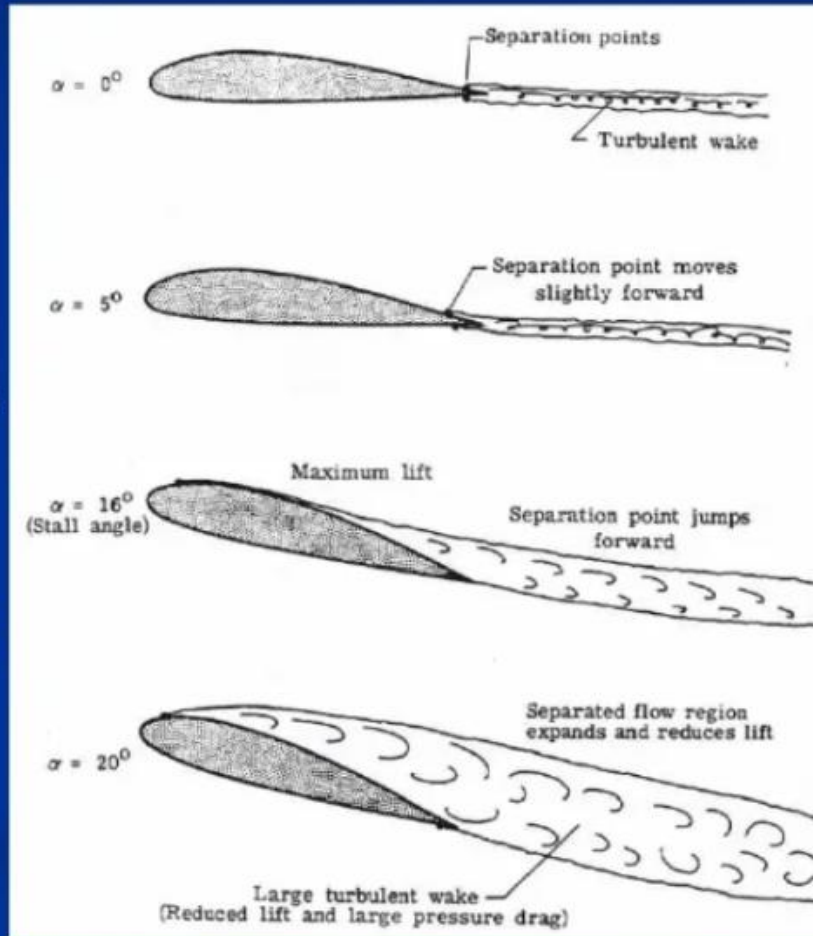


Theory

Lift, Drag and Angle of Attack



Stall Angle



$$\text{Reynolds Number} = \text{Re} = \frac{\rho V c}{\mu} = \frac{\text{Momentum}}{\text{Viscous}}$$

Theory

Direct Method (Force Balance)

$$C_l = \frac{L}{\frac{1}{2}\rho V^2 S} \quad C_d = \frac{D}{\frac{1}{2}\rho V^2 S} \quad \text{Relates lift and drag forces to the velocity}$$

Pressure Distribution (Pressure Ported Airfoil)

$$C_p = \frac{P_{Local} - P_{Stat}}{P_{Dyn}} \quad \text{Relates local pressure on an airfoil to the velocity}$$

$$C_X = \int_{-\frac{y}{c}}^{\frac{y}{c}} (C_{PF} - C_{PA}) d\left(\frac{y}{c}\right)$$

$$C_Y = \int_0^1 (C_{PL} - C_{PU}) d\left(\frac{x}{c}\right)$$

$$C_l = C_Y \cos \alpha - C_X \sin \alpha$$

$$C_d = C_Y \sin \alpha + C_X \cos \alpha$$

Experimental Apparatus

Baylor University Wind Tunnel



24" by 24" Test Section

Test Range: 0 – 150 ft/s

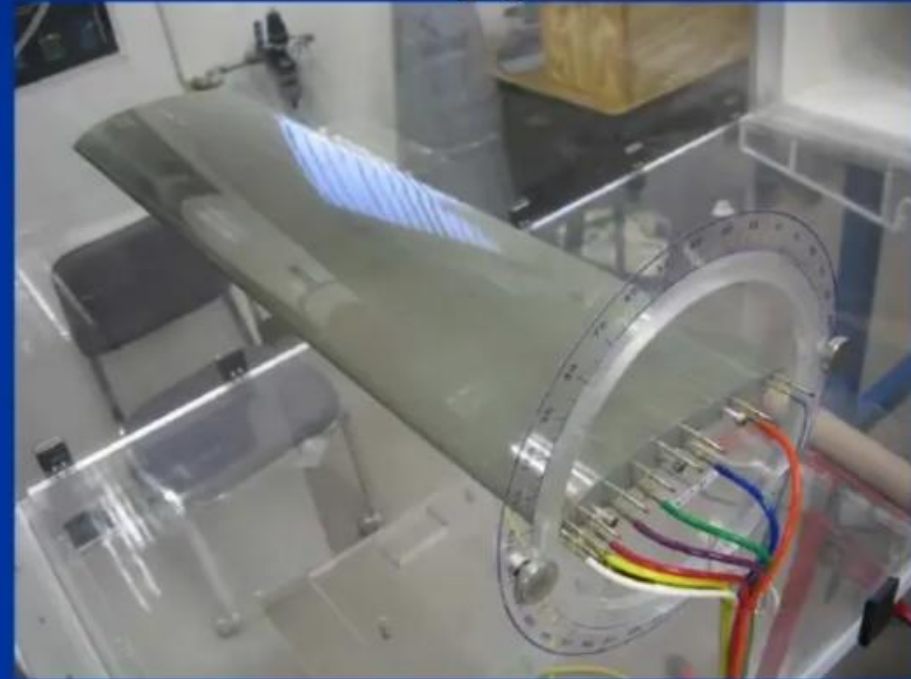
Open loop tunnel

Experimental Apparatus

Force Balance



Pressure Tapped Airfoil



-8 to 20 Degrees

Both NACA 4412 airfoils
are 24" wide with a 6"
chord length

18 pressure ports
-18 to 20 Degrees

Experimental Comparison

NACA

- $Re = 3,000,000$
- 54 pressure ports
- Variable density wind tunnel
- 24" chord length

Baylor University

- $Re = 150,000$
- 18 pressure ports
- Constant density wind tunnel
- 6" chord length

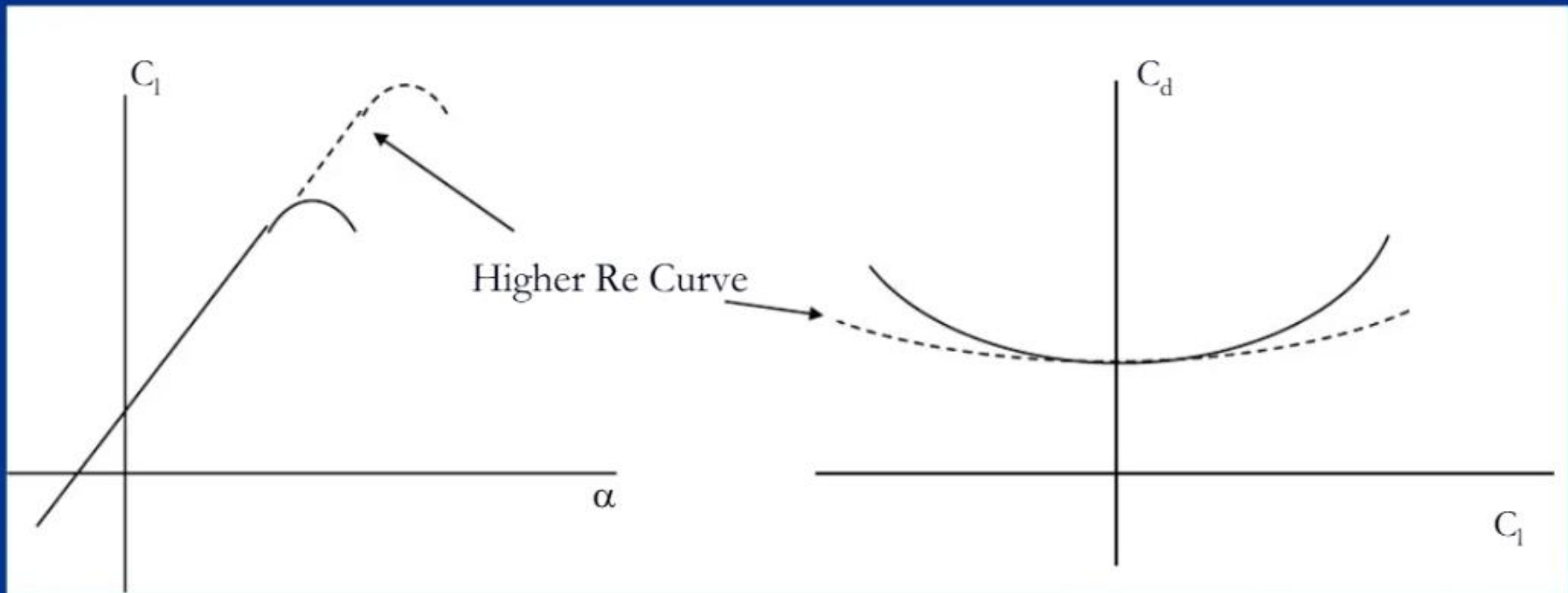
Results

- Stall angle
 - 11 degrees for 150,000 Re (Baylor)
 - 15 degrees for 3,000,000 Re (NACA)
- Lift coefficient agrees within 2% of NACA published data
- Noticeable inaccuracies in drag coefficient data from the pressure ported airfoil
- Drag coefficient is Re dependent

Aerodynamic Curves

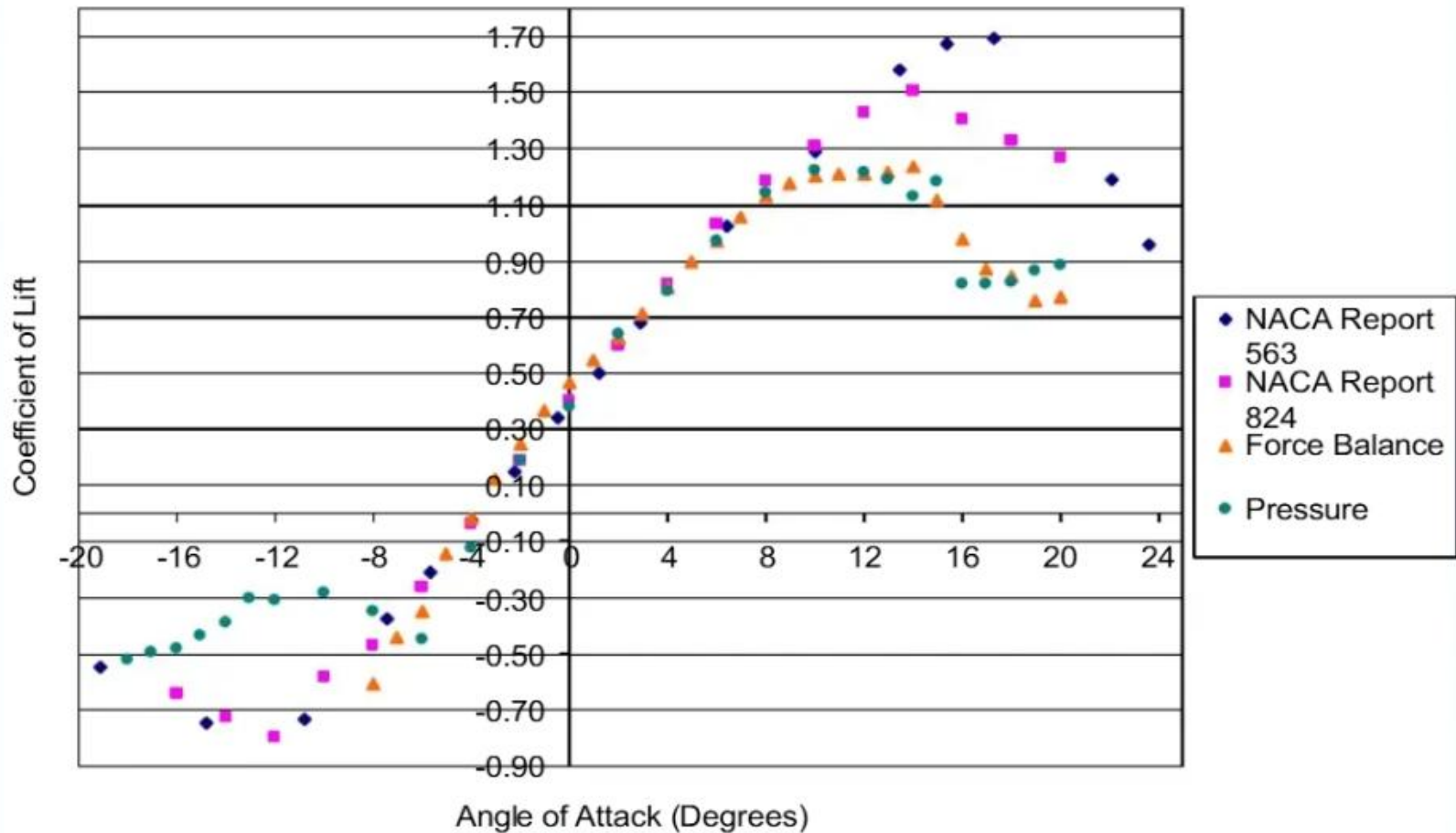
Lift Curve

Drag Curve



Lift Curve

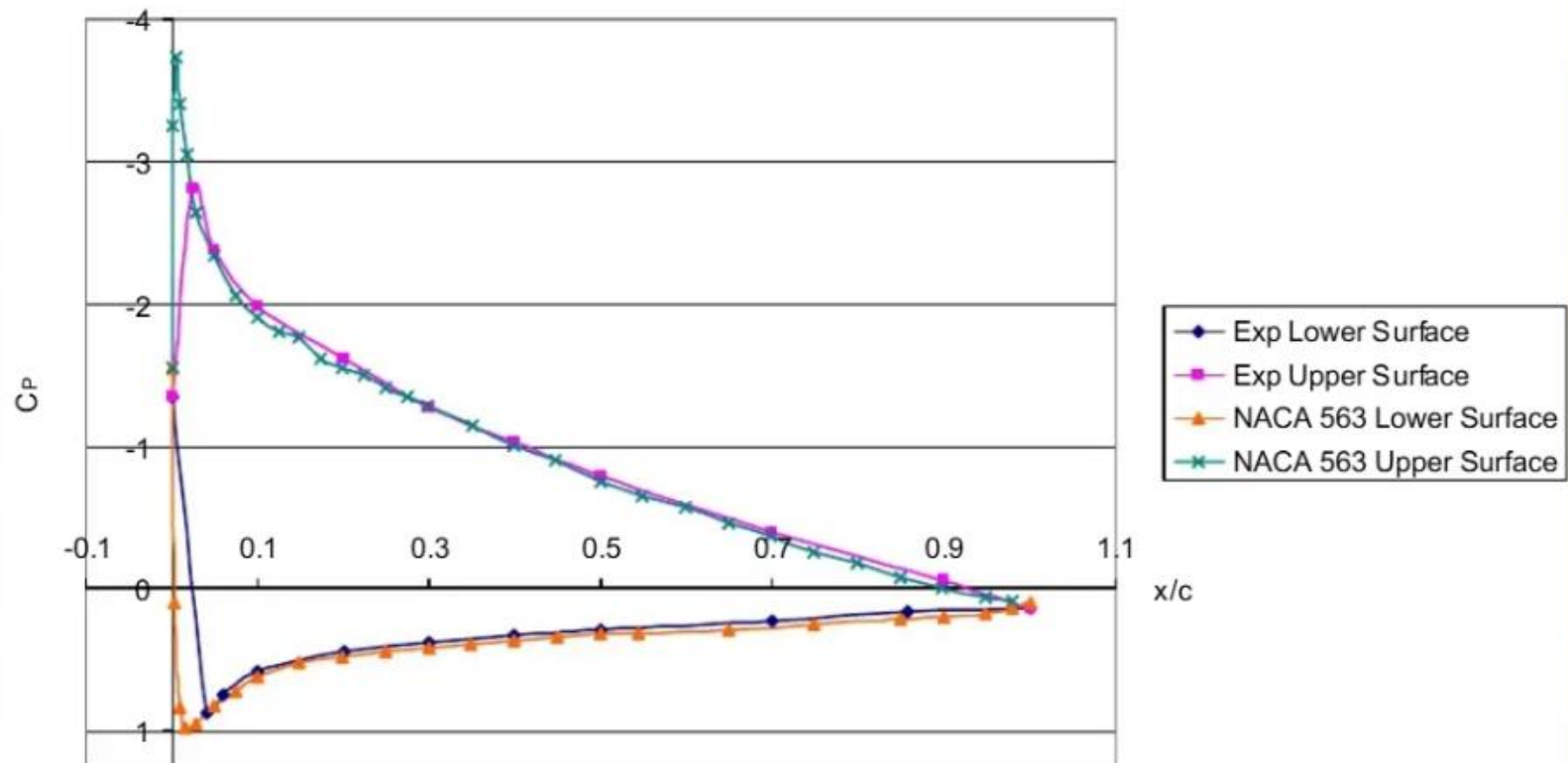
$C_l \text{ v } \alpha$



Lift Pressure Distribution

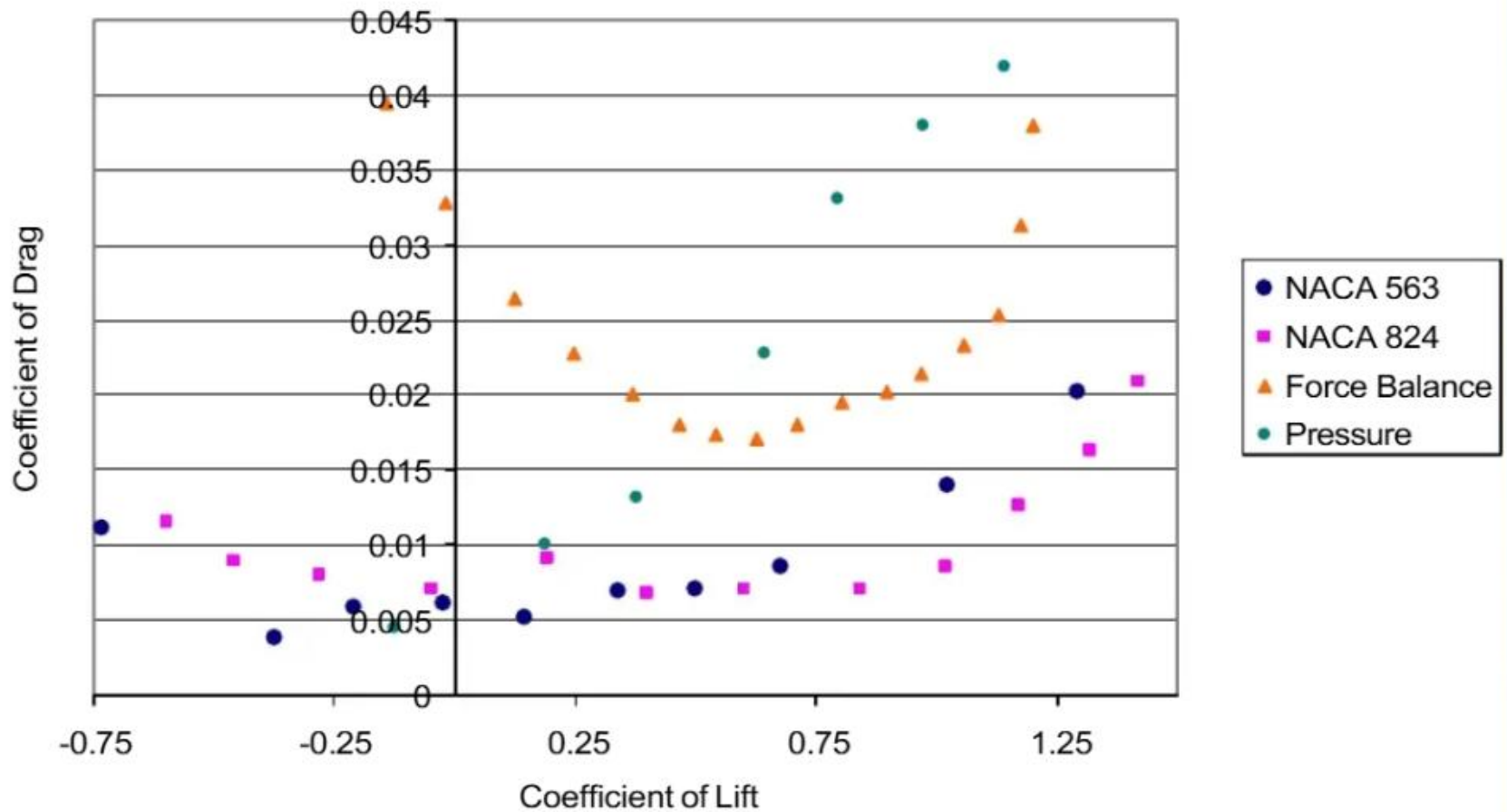
10 degrees

C_p vs. x/c



Drag Curve

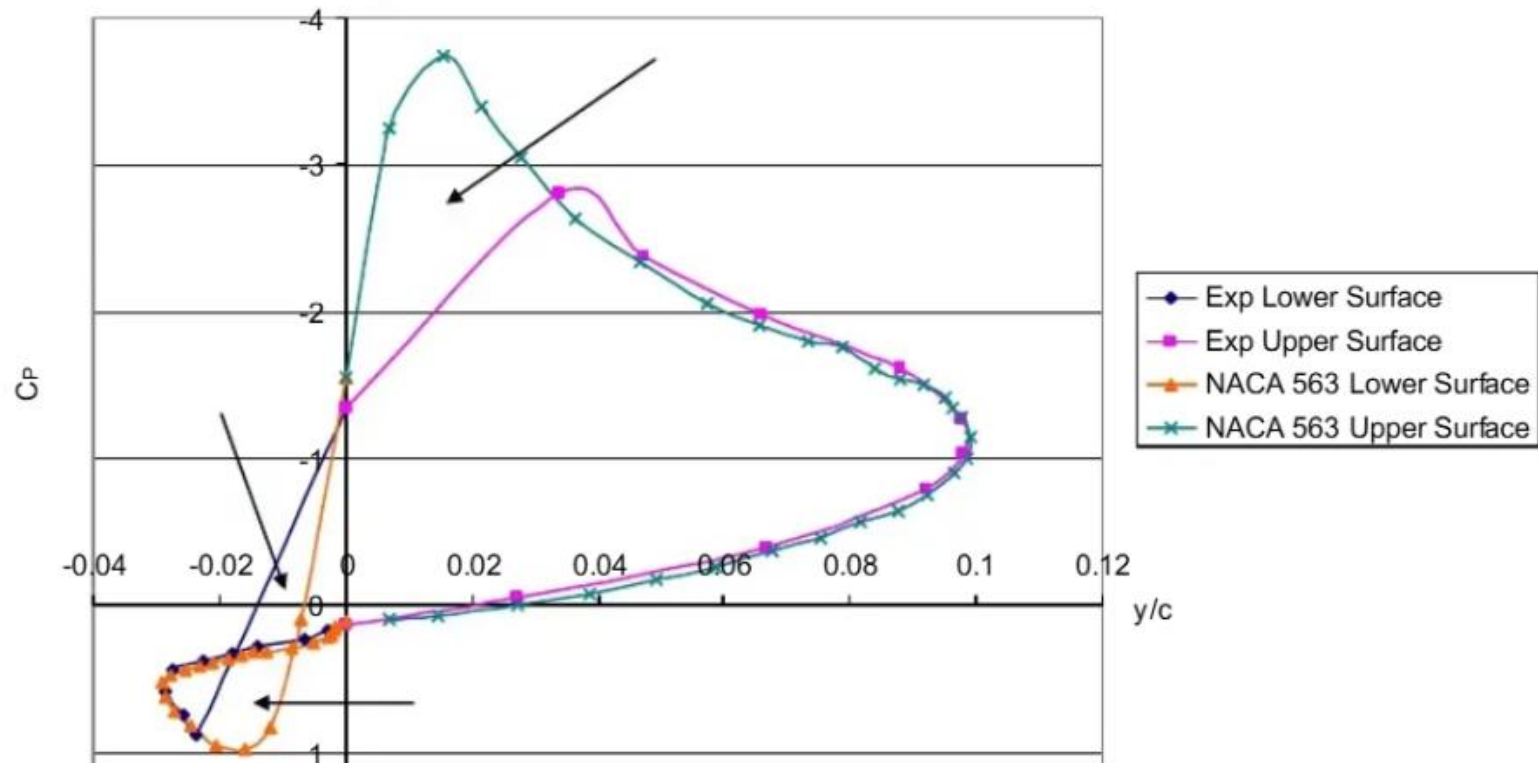
C_D v C_L



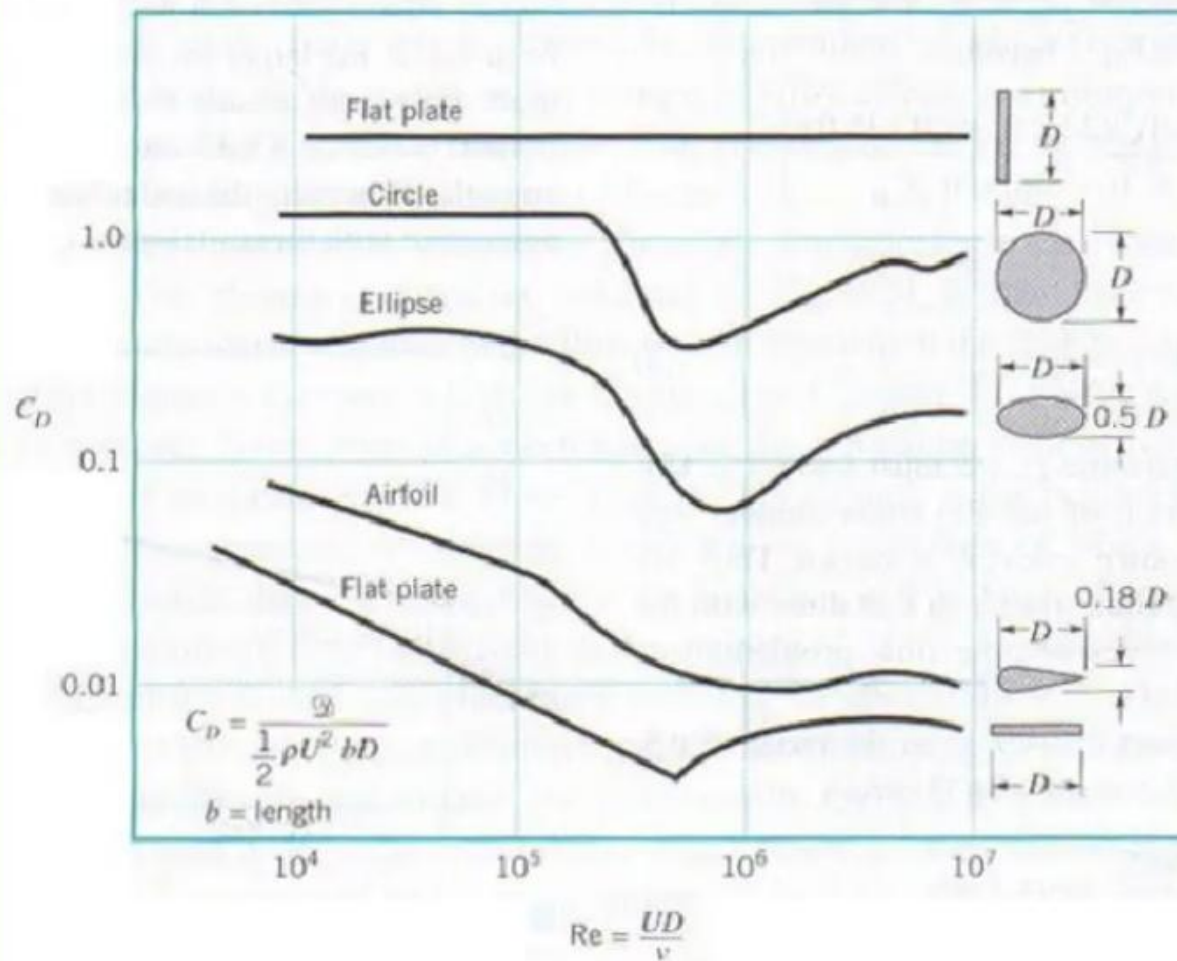
Drag Pressure Distribution

10 degrees

C_p vs. y/c



C_D vs. Reynolds Number



Summary

■ Objectives

- Study airflow over an airfoil
- Resolve discrepancy in previous wind tunnel data
- Compare wind tunnel data

■ Results

- Stall angle is a function of the Reynolds number
- Lift coefficient relates closely to published data
- Insufficient pressure ports to accurately map the pressure distribution for drag coefficient
- Drag coefficient highly dependent on Reynolds number

Recommendations

- Further experiments
 - NACA 0012 (Double the pressure ports)
 - Utilize Baylor's 3D printer
- Develop lift and drag curves for future experiments to reference

Questions

